

Evaluation of a Machine-Learning Classifier for Keratoconus Detection Based on Scheimpflug Tomography

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Purpose: To evaluate the performance of a support vector machine algorithm that automatically and objectively identifies corneal patterns based on a combination of 22 parameters obtained from Pentacam measurements and to compare this method with other known keratoconus (KC) classification methods.

Methods: Pentacam data from 860 eyes were included in the study and divided into 5 groups: 454 KC, 67 forme fruste (FF), 28 astigmatic, 117 after refractive surgery (PR), and 194 normal eyes (N). Twenty-two parameters were used for classification using a support vector machine algorithm developed in Weka, a machine-learning computer software. The cross-validation accuracy for 3 different classification tasks (KC vs. N, FF vs. N and all 5 groups) was calculated and compared with other known classification methods.

Results: The accuracy achieved in the KC versus N discrimination task was 98.9%, with 99.1% sensitivity and 98.5% specificity for KC detection. The accuracy in the FF versus N task was 93.1%, with 79.1% sensitivity and 97.9% specificity for the FF discrimination. Finally, for the 5-groups classification, the accuracy was 88.8%, with a weighted average sensitivity of 89.0% and specificity of 95.2%.

Conclusions: Despite using the strictest definition for FF KC, the present study obtained comparable or better results than the single-parameter methods and indices reported in the literature. In some cases, direct comparisons with the literature were not possible because of differences in the compositions and definitions of the study groups, especially the FF KC.

Key Words: cornea, keratoconus, pentacam, detection, classification (*Cornea* 2016;35:827–832)

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Keratoconus (KC) is a bilateral progressive condition in which the cornea assumes a conical shape as a result of noninflammatory thinning and protrusion,¹ resulting in irregular astigmatism (AST) and decreased vision. The estimated prevalence of KC is 1/1150 (ie, 0.87% of the population),² and it is associated with genetic and environmental factors such as eye rubbing.³ It typically starts at puberty, and it may progress until the fourth decade of life or later.

Until relatively recently, treatment options ranged from contact lenses and implantable ring segments for mild to moderate KC to partial or full-thickness corneal transplants for severe KC.⁴ Corneal collagen crosslinking is a newer treatment for moderate KC that can slow down, or even halt its progression.⁵ Long-term stabilization has been documented in the literature with follow-up periods of up to 10 years.⁶ It is therefore essential that the disease is detected in an early stage, so crosslinking can be applied, preventing a major deformation of the cornea and potentially the need for a corneal transplant.

Advanced degrees of KC are not typically a diagnostic challenge because using a classic keratometer⁷ the highly irregular shape of the cornea would be noticeable. Mild KC, however, presents larger diagnostic challenges that were addressed in the 1980s by the introduction of computerized videokeratography and later by Scheimpflug tomography. These devices provided considerably more information on the shape of the anterior corneal surface. Based on videokeratography maps, the typical characteristics of KC were established as a zone of increased corneal power, clear inferior–superior (I-S) corneal power asymmetry, and skewing of the steepest radial axes above and below the horizontal meridian. Scheimpflug tomography adds the imaging of the posterior corneal surface, producing also a corneal thickness map.

The availability of such a large number of parameters led to the introduction of specifically designed indices that discriminate KC from normal (N) eyes.^{8–14} These were very successful in detecting early to moderate degrees of KC, but considerably less sensitive to very early “subclinical” and “forme fruste” (FF) forms of KC. FF is particularly difficult to detect, as it is relatively stable, and may only decompensate after the cornea is structurally weakened by laser refractive surgery. It was first described by Amsler¹⁵ as an incomplete, abortive, or unusual form of KC with subtle topographic characteristics that remain below the threshold for KC suspects. This description is ambiguous, however, and there

are currently no clear criteria to discriminate FF KC from either N corneas or asymptomatic subclinical KC.

In the present study, we aim to detect asymptomatic FF eyes in an automatic and objective way before they show any sign visible through routine clinical tests. This would be of practical benefit for the prevention of postrefractive (PR) ectasia in corneal refractive surgery, which may still be challenging at times because of the strong clinical similarities between subclinical or FF KC and N eyes. It is therefore of potential clinical benefit that reliable, highly sensitive screening tests for asymptomatic subclinical and FF keratoconic eyes become available. We intend to develop such a screening tool based on a machine-learning algorithm that detects different corneal patterns based on Scheimpflug tomography. The tool will be able to assist the ophthalmologist with the differential diagnosis of patients with KC, which must be considered alongside other clinical data, such as family history, refraction, slit-lamp examination, and evolution of the corneal shape. For this purpose, a support vector machine (SVM) algorithm is assessed for its ability to automatically classify corneas into N, regular AST, PR surgery, FF KC and KC. These results are also compared with the automated classification system of the Pentacam.

SUBJECTS AND METHODS

Subjects

This study was conducted retrospectively at the Antwerp University Hospital using data of 860 eyes (670 subjects) with a mean age of 33.9 ± 9.5 years (range, 17–52 years). Exclusion criteria included any systemic disease, a history of ocular surgery (except for laser refractive surgery in the PR group), crosslinking, individuals with a degree of ametropia more than ± 10 D (for the N group), and very advanced KC with corneal scarring. Rigid contact lens wearers were asked to remove their lenses at least 2 days before the study. Written informed consent was obtained from all participants (or caregivers in case of subjects younger than 18 years) for their clinical records to be used in this study, and approval was obtained from the Ethical Committee of the Antwerp University Hospital.

An experienced KC specialist (C.K.) and an experienced optometrist (I.R.H.) together classified the eyes into 5 groups using criteria based on patient history and anterior and posterior corneal topography maps. Note that KC suspect cases (ie, eyes that the investigators could not reliably categorize as either N or KC based on the Pentacam or slit-lamp examination) were removed from the analysis, as these could not be properly categorized for training purposes.

- Normal (N): Eyes that had no systemic or ocular diseases, prior ocular surgery, or regular AST > 1.5 D.
- Astigmatism (AST): Symmetric and straight bow-tie pattern on the anterior keratometry with AST greater than 1.5 D.
- After refractive surgery (PR): Subjects who had undergone laser-assisted subepithelial keratectomy for myopia, hyperopia, and/or AST correction. The spherical corrections ranged between -1 and -6 D, and cylindrical corrections ranged between -1 and -3 D.

- Forme fruste (FF): Nonsymptomatic Pentacam tomography and elevation, but fellow eye has manifest KC.
- Keratoconus (KC): Diagnosed as mild to moderate KC taking into account the presence of a zone of increased power with abnormal thinning and elevation in the anterior and posterior surfaces combined with patient and family history, changes in refraction and slit-lamp examination.

A description of the most important biometric parameters of each group is given in Table 1.

Methods

Scheimpflug tomography was performed for all eyes by an experienced operator using the Pentacam HR, software version 1.20 r02 (Oculus, Wetzlar, Germany). Based on these examinations, 22 topographic and tomographic parameters were selected from the comma-separated value files in the root folder of the Pentacam program, which are automatically updated each time an examination is either performed or opened. The chosen parameters are available in the most basic version of the Pentacam software (to the best of our knowledge) and include aspects of the corneal curvature, eccentricity, anterior chamber, corneal volume, and pachymetry.

Data were analyzed with Weka 3.7 (University of Waikato, New Zealand). Weka¹⁶ is an open-source suite of machine-learning algorithms with a graphical interface that supports different data mining tasks, including data preprocessing before classification. The algorithm used for classification was SVM.¹⁷ Finally, accuracy was estimated using a 10-fold cross-validation (CV) method on the full data set.¹⁸

Data Preprocessing

The data set used in this work was reviewed for improper and unrealistic values using outlier detection and histograms to ensure that no out-of-range (unrealistic) values were used for either training or testing.

Another source of uncertainty is missing values. The SVM in Weka automatically handles this issue by using a mean imputation technique.¹⁹ This approach represents a compromise between a reduced data set with deleted variables or observations because of missing values and a potential bias introduced by the mean imputation method. Finally, a normalization procedure was carried out to bring all variables to a common scale, so that they all fit into a specific range and are suitably ready to be applied to the machine-learning algorithms extracting the correct information.

Dimensionality Reduction

Although the Pentacam software can provide over 1000 parameters, only the ones that are available in the most basic version of the Pentacam software were used to facilitate independent replication of the study. Because this set of variables still includes irrelevant or highly correlated variables that add redundant information, it is useful to reduce its size further to simplify both computation and interpretation of the classifiers. For this purpose, we performed a correlation-based hierarchical clustering.²⁰ The output of this analysis is

TABLE 1. Description of the Study Groups

Group	Eyes (Subjects)	Mean $\pm$ Standard Deviation (Range)		
		Age (yrs)	K_{mean} (D)	Astigm (D)
Normal (N)	194 (194)	36.7 $\pm$ 9.6 (18–50)	43.3 $\pm$ 1.3 (39–46)	0.82 $\pm$ 0.48 (0.2–3.4)
AST	28 (18)	29.7 $\pm$ 8.5 (20–50)	43.4 $\pm$ 2.1 (40–49)	2.63 $\pm$ 0.78 (1.5–4.3)
PR surgery	117 (62)	37.6 $\pm$ 9.5 (21–52)	39.3 $\pm$ 2.9 (29–46)	0.89 $\pm$ 0.70 (0.0–4.5)
FF	67 (67)	31.4 $\pm$ 9.7 (18–50)	42.8 $\pm$ 1.5 (39–47)	0.85 $\pm$ 0.53 (0.1–3.7)
KC	454 (329)	32.5 $\pm$ 8.9 (17–50)	44.8 $\pm$ 2.1 (39–51)	2.72 $\pm$ 1.69 (0.0–7.9)

Group	Mean $\pm$ Standard Deviation (Range)			
	K_{max} (D)	Pachy _{min} (μm)	Pachy _{apex} (μm)	I-S Value (D)
Normal (N)	44.3 $\pm$ 1.4 (41–48)	548 $\pm$ 29 (480–626)	552 $\pm$ 29 (488–632)	0.02 $\pm$ 0.60 (–1.64 to 1.65)
AST	45.2 $\pm$ 2.2 (41–50)	538 $\pm$ 37 (486–606)	542 $\pm$ 36 (487–613)	0.12 $\pm$ 0.62 (–1.07 to 1.18)
PR surgery	44.2 $\pm$ 2.3 (38–50)	487 $\pm$ 53 (356–606)	499 $\pm$ 55 (367–616)	–0.25 $\pm$ 2.07 (–10.31 to 4.64)
FF	43.9 $\pm$ 1.5 (41–47)	529 $\pm$ 29 (458–591)	534 $\pm$ 29 (466–593)	0.59 $\pm$ 0.56 (–0.76 to 1.61)
KC	49.9 $\pm$ 3.1 (41–55)	478 $\pm$ 38 (346–569)	493 $\pm$ 37 (352–588)	4.74 $\pm$ 2.68 (–1.32 to 13.24)

Astigm, anterior corneal AST; K_{max} , maximum keratometry; K_{mean} , mean keratometry; Pachy_{min}, minimum pachymetry.

a dendrogram, in which variables are organized along branches according to their degree of correlation with each other. Next, the dendrogram was revised to select only one variable from each branch, which was typically the variable with the highest clinical relevance. This procedure led to a data set of 22 variables, which were radii of curvature in the horizontal and vertical meridians (Rh, Rv), central and peripheral AST (Astig, Asti.Periph), maximum keratometry (K_{max}), vertical location of K_{max} from the corneal apex ($K_{\text{max}Y}$), corneal volume (Cor.Vol.), keratometric power deviation (KPD), eccentricity in superior (Ecc Sup) and inferior (Ecc Inf) meridians, anterior and posterior corneal shape factor (Ecc Front, Ecc Back), difference between the minimum and maximum radii of curvature of the spherical component (SphDiff), average value of all higher order aberrations (Irregul), minimum corneal thickness (Pachy_{min}), vertical position of the thinnest point of the cornea from the corneal apex (Pachy_{minY}), maximum pachymetry progression (RPI_{max}), steep keratometry of the posterior surface (K2B), anterior chamber depth at the apex (ACD), I-S index (I-S Value), sagittal height mean (Sag H_{mean}), and difference in pachymetry between the corneal apex and the thinnest point.

Classifier

All classifications were performed using SVM with a linear kernel, one of the most recent and successful

supervised machine-learning algorithms.²¹ This method provides a linear decision boundary for binary classification problems.²²

Classification Evaluation Criteria

The quality of classification by each predictive model was assessed through several parameters based on the number of true-positive (tp), true-negative (tn), false-positive (fp), and false-negative (fn) cases. These parameters can range between 0 and 1 (or between 0% and 100%), with 1 representing the best values and 0 the worst values:

- Accuracy: sum of correct classifications divided by the total number of classifications: $(\text{tp} + \text{tn})/(\text{total number of eyes})$. In a multiclass model, the true negatives (tn) are calculated as the sum of tn predictions from all columns in the confusion matrix. The method used to assess accuracy was a 10-fold CV accuracy procedure²³ on the complete data set.
- Precision: $\text{tp}/(\text{tp} + \text{fp})$
- Sensitivity: $\text{tp}/(\text{tp} + \text{fn})$
- Specificity: $\text{tn}/(\text{tn} + \text{fp})$
- Area under the curve (AUC): the relationship between sensitivity and specificity given by the area under the cumulative distribution function of the detection probability on the y-axis versus the cumulative distribution function of the probability for a false positive on the x-axis, between $\pm \infty$.

TABLE 2. Confusion Matrix (Actual vs. Predicted Classes), CV Accuracy (CV Acc.), Precision (Prec.), Sensitivity (Sens.), Specificity (Spec.), and Area Under the Receiver Operating Characteristics Curve (AUC) of the Support Vector Machine Classification Between the Groups KC and N

Actual Class	Predicted Class		Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	AUC (%)
	KC	N					
KC	454	450	98.9	99.3	99.1	98.5	99.8
N	194	3		97.9	98.5	99.1	99.8

TABLE 3. Confusion Matrix (Actual vs. Predicted Classes), CV Accuracy (CV Acc.), Precision (Prec.), Sensitivity (Sens.), Specificity (Spec.), and Area Under the Receiver Operating Characteristics Curve (AUC) of the Support Vector Machine Classification Between the Groups FF and N

Actual Class		Predicted Class		Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	AUC (%)
		FF	N					
FF	67	53	14	93.10	93.0	79.1	97.9	92.2
N	194	4	190		93.1	97.9	79.1	92.2

RESULTS

To assess the algorithm’s ability to accurately classify corneas in the 5 clinical groups, 3 classification tasks were performed: KC versus N, FF versus N, and a classification including the 5 groups. The accuracy, precision, sensitivity, specificity, and AUC were calculated using the CV method.

KC and N Eyes

The results of the classification task between the KC and N groups are presented in Table 2. The accuracy achieved was 98.9%, and sensitivity and specificity for KC detection were 99.1% and 98.5%, respectively. There are only 7 eyes misclassified (out of 648).

N and FF Eyes

The accuracy is lower in this classification task (93.1%) than in the previous one (Table 3). The sensitivity is 79.1% for FF detection and specificity is 97.9%. The reason for obtaining lower accuracy values in this task is because of the large topographical similarities between N and FF eyes, which make this classification problem much more challenging.

Five-Groups’ Classification: AST, FF, KC, and N and PR Surgery

The global accuracy for the classification task that involved the 5 groups was 88.8%, with a weighted average sensitivity of 89.0% and specificity of 95.2% (Table 4). The sensitivity of the FF group detection is here especially low, which is because of, as mentioned before, the large similarities between FF and N surgeries according to our study groups’ descriptions.

Clinically Relevant Misclassifications

Not all misclassifications in Tables 2–4 are clinically relevant, because a false positive (eg, N eye classified as KC) is far less problematic than a false negative (eg, KC classified as N eye). This issue is especially apparent in the 5-group comparison because of the high number of possible misclassifications, leading to an overestimation of the clinical impact. To mitigate this issue, we introduced the definition of clinically “dangerous” misclassifications as KC or FF misclassified as N, and clinically “safe” misclassifications as N classified as KC or FF. Based on these definitions, we found that clinically safe classifications were up to 6.4% higher than the regular CV accuracy (Table 5).

Pentacam KC Classification System

The Pentacam has a KC classification system based on the front shape of the cornea, ranging from “Pre/early stage” to “stage 4,” which is an adaptation of the Amsler–Krumeich classification system. To compare our model with this scale, we analyzed the outcome of the data from Table 4 where the 5-groups classification is shown, and checked it with the Pentacam’s KC score for each subject. We equated the “possible” group of the Pentacam scale as “FF” in our model, and all degrees of KC (1–4) were merged into the group “KC.”

Table 6 shows the confusion matrices of the classification obtained with both methods. The first column (manual classification) shows the groups and number of subjects per group that were used to create this classification model. As observed in the table, the Pentacam KC classification tends to classify too many eyes into the N category, even though they might be FF, AST, PR, and even KC. For instance, there are 58 (12.8% misclassification rate) KC eyes classified as N, and 63 of 66 FF eyes were classified as N (94% misclassification rate), 65.8% of the corneal surgery eyes and 100% of the

TABLE 4. Confusion Matrix (Actual vs. Predicted Classes), CV Accuracy (CV Acc.), Precision (Prec.), Sensitivity (Sens.), Specificity (Spec.) and Area Under the Receiver Operating Characteristics Curve (AUC) of the Support Vector Machine Classification Between the 5 Groups (AST, FF, KC, N, and PR)

Actual Class		Predicted Class					CV Acc. (%)	Prec. (%)	Sens. (%)	Spec. (%)	AUC (%)
		AST	FF	KC	N	PR					
AST	28	20	2	3	3	0	88.80	74.1	71.4	99.2	98.2
FF	67	3	25	15	23	1		52.1	37.3	98.0	92.5
KC	454	2	10	437	2	3		94.2	96.3	93.6	99.0
N	194	2	9	2	179	2		84.4	92.3	95.1	98.1
PR	117	0	2	7	5	103		94.5	88.8	99.2	95.8

TABLE 5. Improvement in the CV Accuracy When Only the Problematic Classifications Are Taken Into Account in the 3 Analyses Performed in Weka With a SVM Using 22 Variables

Analysis	CV Acc. (%)	Clinically Safe Acc. (%)	Improvement (%)
KC-N	99.1	99.4	0.3
FF-N	93.2	94.6	1.4
N-FF-AST-KC-PR	88.8	95.2	6.4

Acc, accuracy.

ASTs were classified as N. The Pentacam classification system could underestimate the topographical signs of cone-like ectasia, classifying too many eyes from different classes as N. This may be because of the fact that the Pentacam only takes the anterior surface into account for the classification, as mentioned in its documentation.

DISCUSSION

Given the importance of screening patients before refractive surgery, there is great interest in detecting the earliest form of KC. Such detection methods should preferably be objective and could be achieved using machine-learning methods, which are specially designed computer algorithms that use a combination of parameters to automatically and objectively classify eyes into different categories.

The results of this study show that a machine-learning technique can be applied to a large amount of tomographic parameters provided by a Scheimpflug device and can accurately classify eyes as belonging to one of the different given categories (KC, FF, AST, PR, and N).

The parameters used by the classification algorithm agree quite well with the recently published consensus paper on KC,⁴ which stated that abnormal posterior ectasia, abnormal corneal thickness distribution, and corneal thinning are mandatory findings for KC detection. In fact, this applies to 7 out of 22 parameters, with the other 15 being keratometry-related parameters that have been traditionally used in KC detection.

There are a number of studies in the literature successfully distinguishing between keratoconic and N corneas using combinations of parameters or a single, specifically designed

index, for instance, the KISA% index by Rabinowitz,⁹ with 99.6% accuracy; the Klyce/Maeda keratoconus Index by Maeda and Klyce,⁸ with 90% accuracy and sensitivity; and the Cone Location and Magnitude Index by Mahmoud et al¹⁰ with 100% sensitivity. Overall, these achievements were similar to the values achieved in this study (98.9% accuracy and 98.8% sensitivity; see Table, Supplemental Digital Content 1, <http://links.lww.com/ICO/A406>).

Discriminating between N and subclinical or FF KC is considerably more difficult because of clinical similarities between both groups. This causes descriptive values to overlap, leading to difficulties in performing a differential analysis based on a single parameter. There have been several studies looking at a direct comparison between FF or subclinical KC and N eyes, while providing the performances' values of the classification (see Table, Supplemental Digital Content 2, <http://links.lww.com/ICO/A406>).

The most sensitive parameters to date for early KC identification were reported by Rabinowitz et al,⁸ which included a combination of videokeratography, wavefront analysis, and optical coherence tomography. They reported an accuracy of 97.3% with their newly derived PA/I-S index. However, they defined FF as corneas that showed skewed radial axes, which is a first clinical sign of KC. In our study, on the other hand, FF was defined as the nonsymptomatic fellow eye of a keratoconic eye, and therefore, the results cannot be directly compared with the findings of Rabinowitz. Furthermore, as they state in their discussion, one of the limitations of the study is that they used the same group of subjects for training and for testing, which could lead to artificially higher accuracy results. In fact, if the same methods were applied in our model, we would have obtained an accuracy of 100%.

The study by Arbelaez et al¹³ is a thorough study and, to our knowledge, is the first one that used an approach similar to ours, achieving very good results for subclinical KC detection (92% of sensitivity; see Table, Supplemental Digital Content 3, <http://links.lww.com/ICO/A406>). However, their subclinical group included KC-like ectasia characteristics. The same limitation is also encountered in the studies of the Uçakhan⁹ and Smajda¹⁰ groups. The studies by Ruiseñor-Vázquez²⁴ and Reddy²⁵ considered early KC instead of FF KC. To our knowledge, there are only 2 studies conducted by Saad and Gatinel¹² and 1 reported by Gi Hyun et al that are comparable

TABLE 6. Confusion Matrices (Manually vs. Automatically Classified) of the Pentacam Classification System and SVM in Weka

Manual Classification (*)	Classified as									
	Pentacam					Weka SVM				
	N	Poss	Abn	KC	C. Surg	N	FF	AST	KC	C. Surg
N (194)	131	0	0	62	1	170	12	2	3	7
FF (67)	63	4	0	0	0	23	25	3	15	1
AST (28)	28	0	0	0	0	3	2	20	3	0
KC (454)	58	17	0	378	0	2	10	2	437	3
C. Surg (117)	77	1	0	2	37	5	2	0	7	103

*Number of subjects per group.

Abn, Abnormal; C. Surg, corneal surgery; Poss, possible KC.

with ours, because they defined the FF group using a method similar to ours. Saad and Gatinel presented good accuracy results in FF-N discrimination tasks (92% and 93% for each of the studies versus 93.1% in the present study), by using the classification output of an artificial intelligence system included in the OPD-Scan (Corneal Navigator; Nidek Co., Ltd.). Gi Hyun et al²⁶ found no differences in keratometry between the groups N (34 eyes) and FF (14 eyes), but they did in asymmetry parameters such as I-S and ST-IN (superotemporal–inferonasal asymmetry), index of vertical asymmetry, and index of height decentration, reporting an AUC of 0.809 for ST-IN. This is lower than the AUC we obtained in this study using a combination of 22 parameters (0.922).

Another index that facilitates early diagnosis of KC versus N eyes as a screening tool for refractive surgery is the Belin/Ambrosio Enhanced Ectasia Display (BAD D), a feature of the Pentacam that combines pachymetry progression with elevation data.¹⁴ However, either method has yet to prove its value in detecting FF KC without any topographical sign (see Table, Supplemental Digital Content 2, <http://links.lww.com/ICO/A406> for a summary of the different studies).

In conclusion, this study reports the development of a screening tool based on machine-learning algorithms that automatically and objectively detects different corneal patterns based on Scheimpflug tomography. The tool can be implemented in the Pentacam as an add-on program to assist the ophthalmologist with the rapid screening of his patients. The test, however, should not be considered as a stand-alone diagnostic tool but as an additional tool alongside other clinical data such as family history, refraction, slit-lamp examination, and evolution of the corneal shape. Although we hope to increase the capacity to detect corneas suspicious for KC, the challenge for the future is to increase the specificity of the diagnosis in asymptomatic subclinical patients and halt disease development and progression.²⁷

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